

Flexibility

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Benefits of Flexibility



- Increased production & Retention of lubricants between connective tissue
- Reduction in severity and frequency of muscular injuries
- Slows down loss of joint mobility due to aging
- Aids in developing & maintaining proper posture, reducing potential back pain
- Increased performance in specific athletics
- May prevent, delay and/or lessen pain associated with arthritis
- Improves balance, stability and co-ordination
- Reduction in stiffness, tightness and muscular tension
- Better neuromuscular coordination.

4 Methods of Stretching

- Ballistic- Uses speed & force to produce rapid, repetitive bouncing or jerking movements
- Dynamic- Safe form of Ballistic- Gradual, controlled movement through full range of motion
- Static- Slowly moving into an extended position, holding for a period of time, then returning to resting length
- Proprioceptive Neuromuscular Facilitation (PNF)- Contract, relax, contract. Time consuming and requires assistance



F.I.T.T. FLEXIBILITY .



- ❖ **F**requency- 2-3x/wk ideal 5-7
- ❖ **I**ntensity- Stretch to mild discomfort
- ❖ **T**ime- Hold for 15-30sec/2-4reps
- ❖ **T**ype- Static, dynamic PNF

EFFECTS OF WARM UP

- Increased heart rate
- Increased blood flow, stroke volume, o₂ exchange
- Increased muscle & blood temperature
- Increased speed & strength of muscle contractions
- Decreased muscle soreness, muscle/tendon strains & tears
- Increased ROM
- Increased muscle, tendon & ligament elasticity
- Decreased muscle tension
- Increased physiological preparedness

EFFECTS OF COOL DOWN

- Aids in circulatory & metabolic exercise rates returning to normal levels
- Decreases muscle cramps
- Decreases muscle fatigue
- Increases tendon, ligament, and muscle elasticity
- Decreases post- exercise muscle soreness
- Decreases muscle stiffness
- Decreases risk of dizziness & fainting
- Prevents a pooling of blood in the veins
- Aids in dispersement of lactic acid
- Promotes relaxation of the body & mind
- Decreases body temperature

Fitness & sports related injuries, Safety, Prevention

- ACUTE- Single event, damaging a part of musculoskeletal system.
Response: Immediate pain/swelling, immovable, non weight bearing, pop/snap, spasm, discoloration, bleeding
 - CHRONIC- Repetitive forces injuring a structure over time.
 - Overuse treatment: Rest, non weight bearing, surgery
- Traumatic Brain Injuries
 - Environmental Conditions
 - Sudden Cardiac Arrest
 - Exercise induced Asthma



Four Phases of Healing

